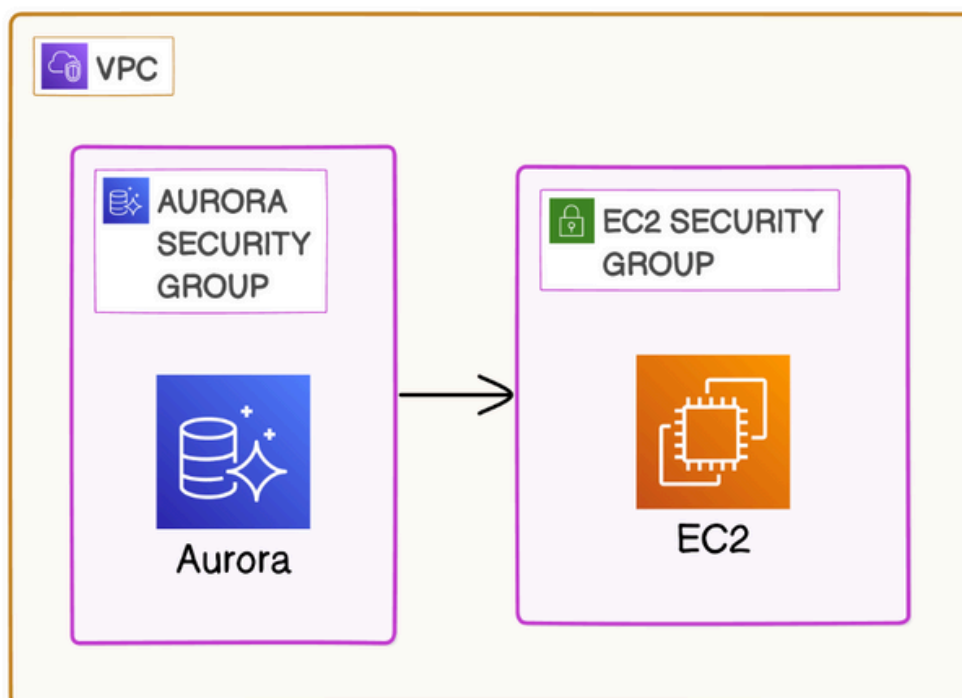


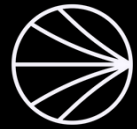


Nikko Cajayon
Community Lead



Cloud Project:
Connecting AWS Aurora Database
to an Elastic Compute Cloud
(EC2 Web-server)



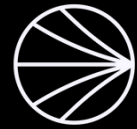


1 INTRODUCTION

Welcome to the second installment of our AWS Database Project series, focused on establishing real-world connectivity for our data infrastructure. In the previous part, we successfully provisioned a highly scalable and resilient Amazon Aurora relational database instance.

This project, "AWS Aurora Database with Elastic Compute Cloud (EC2)," is the critical next step. The goal is to move beyond mere provisioning and create a functional application environment. The Aurora database, by design, is secured within a virtual private cloud (VPC) and isolated from the public internet. This step will demonstrate the professional, secure method for connecting an application server, simulated by an Amazon EC2 instance to the database, ensuring that our data is both accessible to our application code and protected from unauthorized external access.

By the end of this module, you will have a complete, secure backend stack where your application logic runs on a dedicated EC2 server, interacting seamlessly with the powerful Aurora database, ready for deployment and scaling.



TOOLS AND SERVICES

Amazon Aurora

- A fast, highly reliable, and fully managed relational database service compatible with MySQL and PostgreSQL. It is optimized for performance and scales automatically.

Elastic Compute Cloud (EC2)

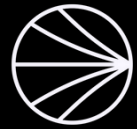
- a service that provides resizable compute capacity (virtual servers) in the cloud. It serves as the application host for connecting to the database.

Relational Databases

- A type of database that stores data in structured tables (rows and columns) and uses keys to define explicit relationships between different data points.

IAM (Identity and Access Management)

- Manages user roles and policies for AWS resources.
- Provides least-privilege access Ensures only authorized users can manage or query the database.

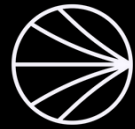


PREPARE AN IDENTITY ACCESS MANAGEMENT (IAM) ACCOUNT

IAM is like a special account you create within your AWS account for different people or applications. It's a way to give them specific permissions to access and manage AWS resources without sharing your main account credentials. Think of it as creating separate logins for your team members, each with limited access to only what they need.

IAM users are crucial for security and access control in AWS. Here's a breakdown:

- **Security:** IAM users prevent you from sharing your root account credentials, which have unrestricted access to everything. This reduces the risk of unauthorized access and potential damage.
- **Granular Permissions:** You can assign specific permissions to each IAM user, controlling which AWS services and resources they can access and what actions they can perform.
- **User Management:** IAM allows you to easily manage users, groups, and roles within your AWS account. You can create groups with common permissions and add users to those groups for simplified administration.
- **Access Keys:** IAM users can be assigned access keys for programmatic access to AWS services via the AWS CLI, SDKs, or APIs. These keys should be managed securely.



2

CREATE AN AURORA MYSQL DATABASE

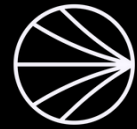
This project outlines a cloud infrastructure build. We will configure an Amazon Aurora database and an EC2 instance, detailing the network connectivity and security measures. The goal is to establish a foundational architecture for a web application.

WHAT IS A RELATIONAL DATABASE?

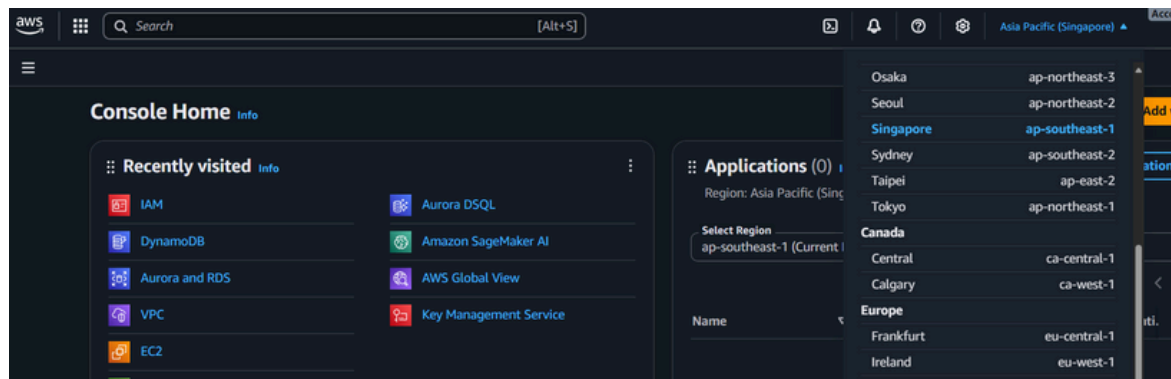
A relational database is a way of organizing information into tables. Think of it like a spreadsheet, where you have rows and columns. The "relational" part means that the different tables can be linked together based on relationships between the data.

A relational database is characterized by the following:

- **Tables:** Data is stored in tables, which are made up of rows and columns. Each row represents a record, and each column represents an attribute of that record.
- **Relationships:** Tables can be linked together using keys. A primary key uniquely identifies a record in a table, while a foreign key establishes a link to a record in another table.
- **SQL:** Relational databases are typically managed using SQL (Structured Query Language), which is used to query, update, and manage the data.
- **ACID Properties:** Relational databases adhere to ACID properties (Atomicity, Consistency, Isolation, Durability), ensuring that transactions are processed reliably.



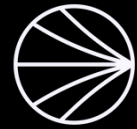
REGION



WHY DOES OUR REGION MATTER?

A Region in AWS refers to a geographical area where Amazon has multiple, physically separated data centers. These data centers are called Availability Zones. Think of a Region as a city, and Availability Zones as different neighborhoods within that city.

- **Geographical Isolation:** Regions are designed to be isolated from each other. This means that a major event in one Region shouldn't affect other Regions.
- **Availability Zones (AZs):** Each Region contains multiple Availability Zones. AZs are designed to be isolated from failures in other AZs.
- **Low Latency:** AWS allows you to choose the Region closest to your users, reducing latency and improving performance.
- **Data Sovereignty:** Regions help comply with data governance and legal requirements by keeping data within specific geographical boundaries.



CREATE AN AURORA MYSQL DATABASE

The screenshot shows the 'Create database' interface in the AWS Management Console. At the top, there are two tabs: 'Standard create' (selected) and 'Easy create'. Below this is the 'Engine options' section, which displays a grid of database engine choices. 'Aurora (MySQL Compatible)' is selected. Other options include Aurora (PostgreSQL Compatible), MySQL, PostgreSQL, MariaDB, Microsoft SQL Server, and IBM Db2. At the bottom, the 'Engine version' is set to 'Aurora MySQL 3.08.2 (compatible with MySQL 8.0.39) - default for major version 8.0'. There is also a checkbox for 'Enable RDS Extended Support'.

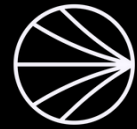
WHAT IS ENGINE TYPE?

In the context of databases refers to the core software that manages and controls the database. Think of it as the operating system for your database, defining how data is stored, retrieved, and processed.

The Engine Type determines the fundamental characteristics and capabilities of the database. Here's a breakdown:

WHAT IS ENGINE VERSION?

The engine version is like choosing the specific version of software that your database will use. For example, Aurora MySQL 3.08.2 (compatible with MySQL 8.0.39) - default for major version 8.0 is a specific version of Aurora that's compatible with a particular MySQL version (8.0.32).



WHY WOULD YOU CHOOSE AURORA FOR YOUR RELATIONAL DATABASE?

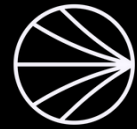
Aurora is a good choice when your application requires high-performance, high-availability, and scalability beyond what standard managed databases (like RDS MySQL/PostgreSQL) can easily offer.

The screenshot shows the AWS Aurora 'Settings' page. It includes sections for 'DB cluster identifier' (with a text input 'my-db-cluster'), 'Credentials Settings' (with 'Master username' set to 'quicksightdatabase'), 'Credentials management' (with 'Self managed' selected), 'Master password' (with a strength indicator 'Very strong'), and 'Cluster storage configuration' (with 'Aurora Standard' selected). The 'Aurora Standard' option is highlighted with a blue border and includes details about cost-effective pricing and I/O charges.

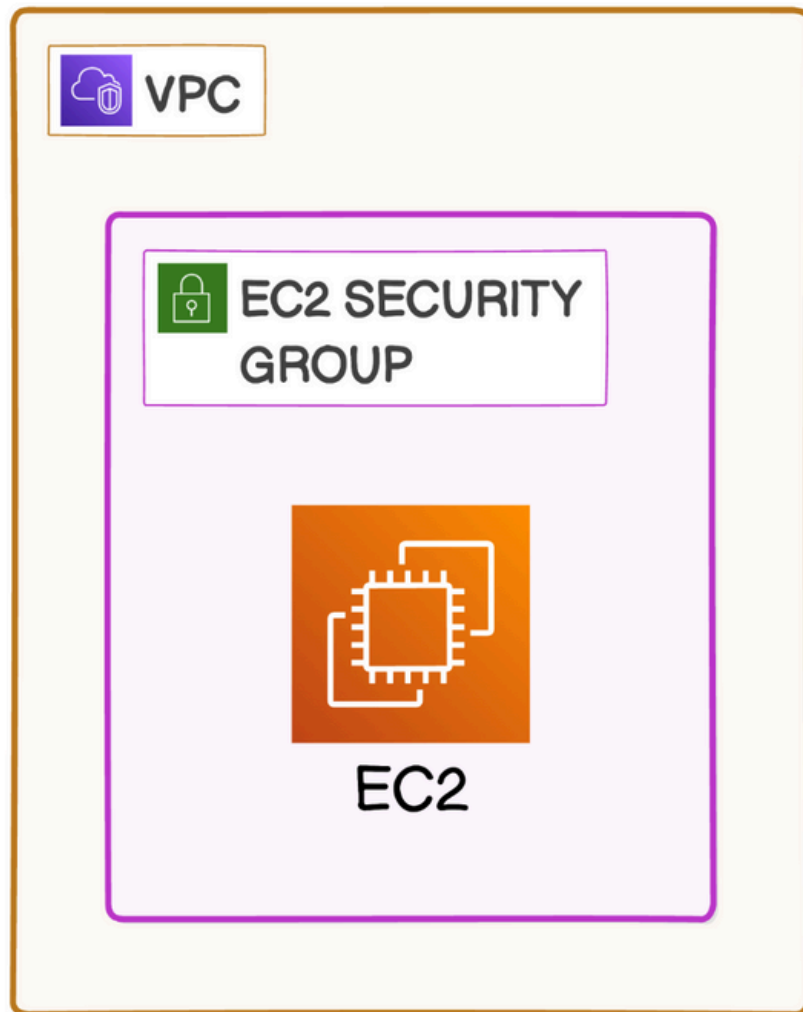
WHAT IS INSTANCE CONFIGURATION?

Instance configuration is where we can select the type of virtual computer that our Aurora database will run on. Just like opening up a Word Document runs on your own computer, we need a computer for our Aurora database to run on too.

The Burstable classes (includes t classes) are cost-effective types of database instances that are best when you have a consistent baseline level of traffic with occasional, random spikes in demand...a sudden "burst" of traffic. By choosing burstable classes, our database can perform well under high traffic, but also save us costs when things are quiet.



BEFORE WE GO ANY FURTHER, HAVE A THINK FOR YOURSELF. DO YOU THINK WE NEED TO CONNECT OUR AURORA DATABASE TO AN EC2 INSTANCE? WHAT IS IT WE'RE TRYING TO ACHIEVE HERE?



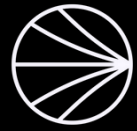
WHY DO WE NEED AN EC2 INSTANCE TO BE A WEB APP SERVER

Any web app needs to run on a computer. But are we going to go buy a physical computer to do this?

No! We're going to use a virtual computer that we rent through AWS. This is exactly what EC2 is here for. Think of an EC2 instance as a rented virtual computer that you use to run your web app. It's the place where all the processing, data handling, and user interactions happen.

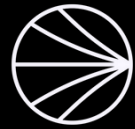


Nikko Cajayon
NextWork Community Lead



WHY DID YOU STOP CREATING YOUR AURORA DATABASE?

Before provisioning the Aurora database, we'll first deploy the EC2 instance, as we'll need its network information (like Security Group ID) during the database setup to configure the secure network connection between the web-app and database.

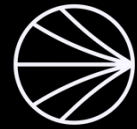


3

LAUNCH AN EC2 INSTANCE

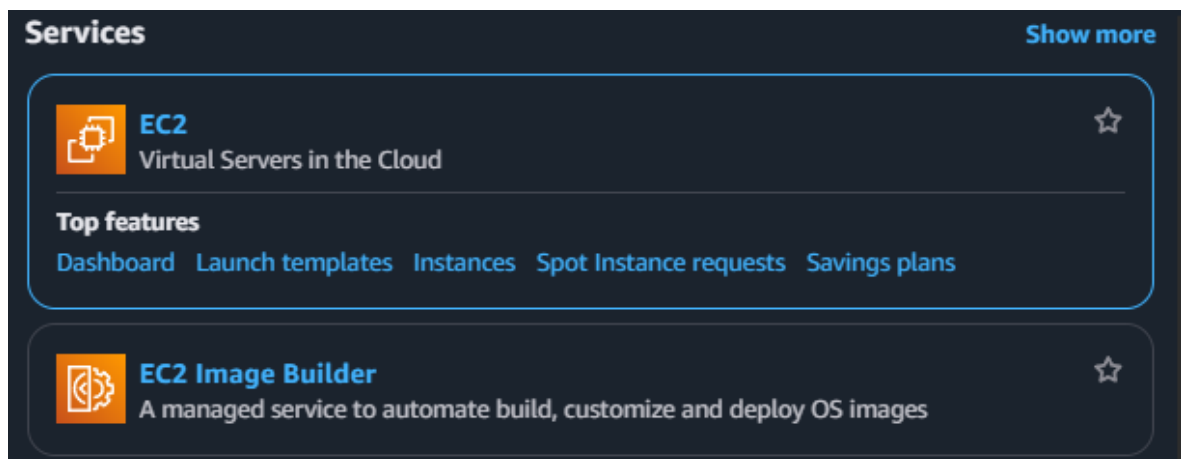
In this step, we will be creating an EC2 instance to host our web app. This virtual server will be provisioned within the VPC and configured with the necessary Security Group rules to control ingress web traffic and establish secure network with DB.





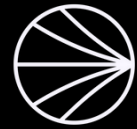
WHAT IS AN ELASTIC COMPUTE CLOUD EC2

An EC2 instance is essentially a virtual computer that you can rent from Amazon Web Services (AWS). Think of it as a computer in the cloud that you have complete control over. You can install software, run applications, and store data on it, just like you would on a physical computer.



An EC2 instance provides scalable computing capacity in the AWS cloud. Here's a more detailed breakdown:

- **Virtual Machine:** It's a virtual machine, meaning it's a software-based emulation of a physical computer.
- **Customizable:** You get to choose the operating system (like Linux or Windows), the amount of processing power (CPU), memory (RAM), and storage you need.



WHAT IS AN ELASTIC COMPUTE CLOUD EC2

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-088d74defe9802f14 (64-bit (x86), uefi-preferred) / ami-03ab2130009e52d4a (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0224 USD per Hour

On-Demand SUSE base pricing: 0.0132 USD per Hour

On-Demand Linux base pricing: 0.0132 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0167 USD per Hour

On-Demand RHEL base pricing: 0.042 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

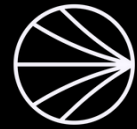
▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

[Create new key pair](#)

A key pair in EC2 is quite literally the keys to access our EC2 instance. We need keys to our EC2 instance if we want to add, change, or update how our EC2 instance is running.



▼ Summary

Number of instances | Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-05fd46f12b86c4a6c

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

What does this Summary panel tell us?

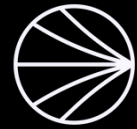
Number of instances - the number of machines you're running.

Software Image (AMI) - the operating system of your virtual machine.

Virtual server type - the CPU, memory, and network performance package you selected.

Firewall - firewalls control the inbound and outbound traffic for your instances. In EC2, we use security groups for this.

Storage - the storage options for your instance.



4

CREATE YOUR AURORA DATABASE

Next, we provision the Aurora DB Cluster. The critical action is configuring the VPC Security Group for the database to explicitly allow inbound network connections from the previously created EC2 Security Group, establishing the secure link.

Connectivity info

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

EC2 instance info
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database to allow the EC2 instance to access the database.

i-04306385827ee7c09
rpg-ec2-instance-web-server

Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Virtual private cloud (VPC) info
Choose the VPC. The VPC defines the virtual networking environment for this DB cluster.

Default VPC (vpc-0391ada3d4750346c)
6 Subnets, 6 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB cluster can use in the VPC that you selected.

☐ Choose existing
Choose existing DB subnet group

☒ Automatic setup
RDS creates a new subnet group for you or reuses an existing subnet group

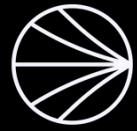
DB subnet group name

WHAT'S THIS 'DATABASE CLUSTER' BUSINESS?

Remember how we said that Aurora is really good for the big jobs? The reason for this is clusters.

A database cluster in Aurora is a group of database copies that work together so your data is always available.

Each cluster consists of a primary instance (where all write operations occur) and multiple read replicas as back-ups. If your database's primary instance fails, one of the replicas can be promoted to primary automatically.



UNDERSTANDING READER VS WRITER INSTANCES

Our **writer** database instance is our primary instance that handles all the "write" operations like INSERT, UPDATE, and DELETE.

Our **reader** database instance is our backup instance that can do very basic operations like SELECT. This is used to get data, but not to add or change data.

WHY SEPARATE READ AND WRITE?

We only want one writer instance at a time so that things stay focused and controlled...but we want multiple reader instances so that we can share the workload for read requests as our database grows and have a backup if our writer instance fails.

WHAT IS AN ENDPOINT?

In general, endpoints are like contact points where data flows in and out. A super popular example is a website URL! When you enter a website's address (like nextwork.org), your browser uses that endpoint to receive data and load the page. For databases, an endpoint is how your app finds a database to ask for data or update it.



Nikko Cajayon
NextWork Community Lead



PROJECT SUMMARY

This project, "Connecting AWS Aurora Database to Elastic Compute Cloud (EC2)," serves as the foundational deployment step in your AWS Database Project series. You have successfully moved beyond provisioning to build a secure, functional backend architecture ready for a web application.

PROJECT SUCCESS

By completing this module, you have achieved a critical architectural milestone: you have established the secure, high-speed internal connection between your application server and your database. This is the professional, best-practice approach for running scalable web applications on AWS.



Nikko Cajayon

NextWork Community Lead
Director for SkillBuilder
CTO of AWS Cloud Club

Here at NextWork We are
Transforming the Way World Learns

One Step Backward, Two Step Forward